

Robotic arm configurations:

For body-and-arm configurations, there are many different combinations possible for a three-degree-of-freedom robot manipulator, comprising any of the five joint types.

Common body-and-arm configurations are as follows.

- 1) Polar coordinate arm configuration
- 2) Cylindrical coordinate arm configuration
- 3) Cartesian coordinate arm configuration
- 4) Jointed arm configuration

1) Polar coordinate arm configuration(RRP):

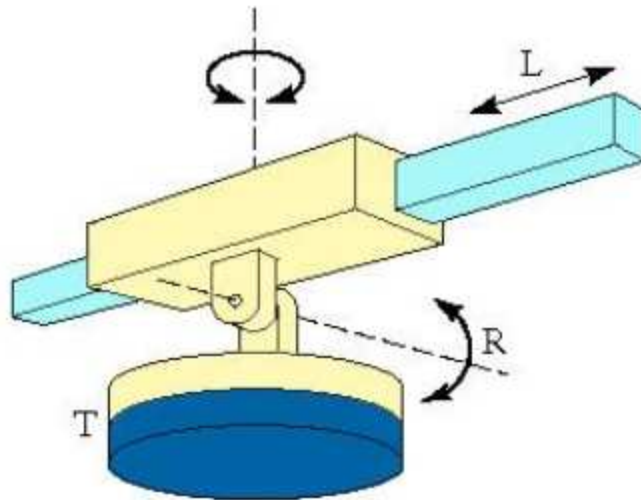


Fig 3: 3-dof polar arm configuration

The polar arm configuration is shown in the fig 3. It consists of a prismatic joint that can be raised or lowered about a horizontal revolute joint. The two links are mounted on a rotating base. These various joints provide the capability of moving the arm endpoint within a partial spherical space. Therefore it is called as Spherical co-ordinate configuration. This configuration allows manipulation of objects on the floor.

Drawbacks:

- i. Low mechanical stiffness
- ii. Complex construction
- iii. Position accuracy decreases with the increasing radial stroke.

Applications: Machining, spray painting

Example: Unimate 2000 series, MAKER 110

2) Cylindrical coordinate arm configuration (RPP):

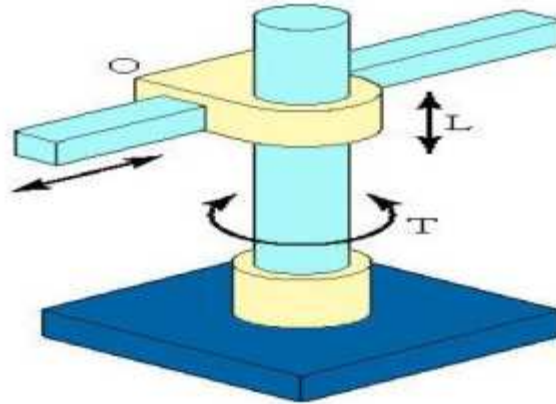


Fig 4: 3-dof cylindrical arm configuration

The cylindrical configuration uses two perpendicular prismatic joints and a revolute joint as shown in fig 4. This configuration uses a vertical column and a slide that can be moved up or down along the column. The robot arm is attached to the slide, so that it can be moved radially with respect to column. By rotating the column, the robot is capable of achieving a workspace that approximates a cylinder. The cylindrical configuration offers good mechanical stiffness.

Drawback: Accuracy decreases as the horizontal stroke increases.

Applications: suitable to access narrow horizontal capabilities, hence used for machine loading operations. **Example:** GMF model M-1A.

3) Cartesian coordinate arm configuration (PPP):

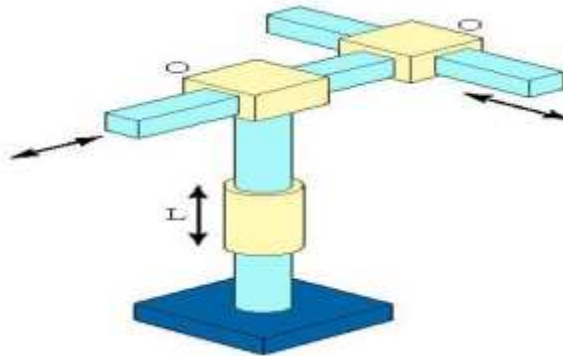


Fig 5: 3-dof Cartesian arm configuration

From fig 5. Cartesian coordinate or rectangular coordinate configuration is constructed by three perpendicular slides, giving only linear motions along the three principal axes. It consists of three prismatic joints. The endpoints of the arm are capable of operating in a cuboidal space. Cartesian arm gives high precision and is easy to program.

Drawbacks:

- o limited manipulatability
- o low dexterity (not able to move quickly and easily)

Applications: use to lift and move heavy loads.

Exa

4) Joint

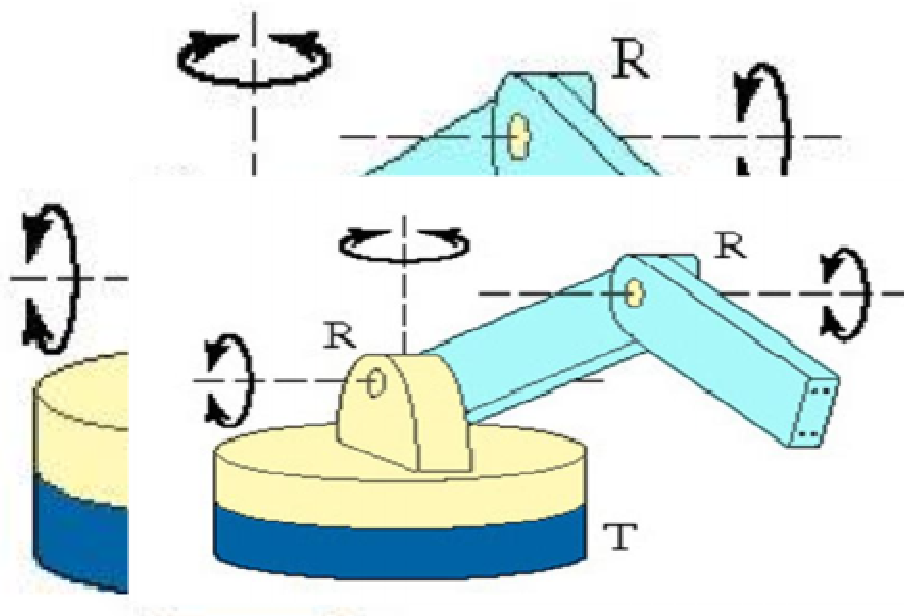


Fig 6. 3-dof jointed arm configuration

From fig 6. jointed arm configurations are similar to that of human arm. It consists of two straight links, corresponding to human 'fore arm' and 'upper arm' with two rotary joint corresponding to the elbow and shoulder joints. These two are mounted on a vertical rotary table corresponding to human waist joint. The work volume is spherical. This structure is the most dexterous one. This configuration is very widely used.

Applications: Arc welding, Spray coating.

Example: SCARA robot (Selective compliance Assembly Robot Arm)

Its full form is 'Selective Compliance Assembly Robot Arm'. It is similar in construction to the jointed-arm robot, except the shoulder and elbow rotational axes are vertical. It means that the arm is very rigid in the vertical direction, but compliant in the horizontal direction.

The SCARA body-and-arm configuration typically does not use a separate wrist assembly. Its usual operative environment is for insertion-type assembly operations where wrist joints are unnecessary. The other four body-and-arm configurations more-or-less follow the wrist-joint configuration by deploying various combinations of rotary joints viz. type R and T

Robot Wrist:

Wrist assembly is attached to end-of-arm. End effectors are attached to wrist assembly. Function of wrist assembly is to orient end effectors. Body-and-arm determines global position of end effector. It has three degrees of freedom:

- **Roll (R)** axis – involves rotation of the wrist mechanism about the arm axis.
- **Pitch (P)** axis – involves up or down rotation of the wrist.
- **Yaw (Y)** axis – involves right or left rotation of the wrist.

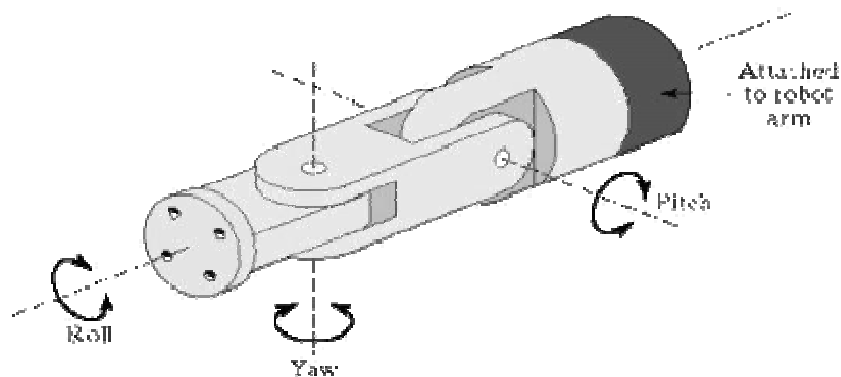


Fig 7: Robotic wrist

Robot wrist assembly consists of either two or three degrees-of-freedom. A typical three-degree-of-freedom wrist joint is depicted in Figure 7; the roll joint is accomplished by use of a T joint; the pitch joint is achieved by recourse to an R joint; and the yaw joint, a right-and-left motion, is gained by deploying a second R joint. Care should be taken to avoid confusing pitch and yaw motions, as both utilize

R

joints.

Degree of freedom:

In mechanics, the degree of freedom (DOF) of a mechanical system is the number of independent parameters that define its configuration. It is the number of parameters that determine the state of a physical system and is important to the analysis of systems of bodies in mechanical engineering, aeronautical engineering, robotics, and structural engineering.

The position and orientation of a rigid body in space is defined by three components of translation and three components of rotation, which means that it has six degrees of freedom.

The motion of a ship at sea has the six degrees of freedom of a rigid body, and is described as shown in fig 8 .

Translation:

1. Moving up and down (heaving);
2. Moving left and right (swaying);
3. Moving forward and backward (surging);

Rotation:

4. ~~Tilts forward and backward~~ (pitching);
5. Swivels left and right (yawing);
6. ~~Pivots side to side~~ (rolling).

From fig 9. The trajectory of an airplane in flight has three degrees of freedom and its attitude along the trajectory has three degrees of freedom, for a total of six degrees of freedom.

Robot work volume:

A space on which a robot can move and operate its wrist end is called as a work volume. It is also referred as the work envelope and work space. For developing a better work volume, some of the physical characteristics of a robot should be considered such as:

- The anatomy of various robots
- The maximum value for moving a robot joint
- The size of the robot components like wrist, arm, and body